

# JEE MAIN REVISION MASTER: SETS AND RELATIONS 2026 4 QUESTIONS 6.67% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 23 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                               | Frequency | Category        |
|---------------|--------------------------------------------|-----------|-----------------|
| Very High     | Kcet Sets And Relations - Core Application | 23 PYQs   | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Kcet Sets And Relations - Core Application (Very High Yield — 23 PYQs)

Core Principle: Key problem pattern in Kcet Sets And Relations.

■■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Let  $R$  be the relation in the set  $\mathbb{N}$  given by  $R = \{(a,b) : a = b - 2, b > 6\}$ . Which of the following is the correct answer?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "If  $n(A) = 2$  and the number of relations from set  $A$  to set  $B$  is 1024, then  $n(B)$  is..." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "If  $A = \{1, 2, 3, 4, \dots, 10\}$ , then the number of non-empty subsets of  $A$  containing only even number is..." [Ans: A]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: KCET 2026 (Q#1)**

Let  $R$  be the relation in the set  $\mathbb{N}$  given by  $R = \{(a, b) : a = b - 2, b > 6\}$ . Which of the following is the correct answer?

- (A)  $(2, 4) \in R$
- (B)  $(3, 8) \in R$
- (C)  $(6, 8) \in R$
- (D)  $(8, 7) \in R$

**Correct Answer: None**

**Solution Analysis:** For a pair  $(a, b)$  to be in the relation  $R$ , it must satisfy both conditions:  $a = b - 2$  and  $b > 6$ . Check the options: For  $(2, 4)$ :  $2 = 4 - 2$  This is true, but  $4 > 6$  is false. So  $(2, 4) \notin R$ . For  $(3, 8)$ : Here  $b = 8$ , so  $b > 6$  is true. But  $b - 2 = 8 - 2 = 6$  So  $a$  should be 6, not 3. Hence  $(3, 8) \notin R$ . For  $(6, 8)$ : Here  $b = 8$ , so  $b > 6$  is true. Also,  $b - 2 = 8 - 2 = 6$  So  $a = 6$ . Both conditions are satisfied. Thus,  $(6, 8) \in R$  So the correct answer is: Option C